Samuel Raju Bethala

B.S. + M.S. Dual Degree in Chemistry
Indian Institute of Technology Bombay

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I am a dual-degree researcher in Chemistry at IIT Bombay with a strong interest in applying artificial intelligence to cancer research, particularly through the development of deep learning models and computer vision pipelines for quantifying tumor dynamics. I am also deeply engaged in computational biophysics, focusing on the integration of molecular dynamics simulations with machine learning frameworks to accelerate structure-based drug discovery.

Education

Indian Institute of Technology Bombay
Bachelor of Science & Master of Science (B.S.+ M.S.) Dual Degree | Department of Chemistry

PUBLICATION

- **Samuel Raju Bethala**, Vanshika, *An Explainable and Comparative Transfer Learning Framework for Brain Tumor Classification from MRI Images.*
  Under Review in Biomedical Signal Processing and Control *medRxiv Preprint*

Research Experience

Temporal Bayesian Optimization of Deep Cell Segmentation for Live-Cell Imaging

M.S. Research Project | TSBL | Guide: Prof. Sandip Kar *(Aug '25 - May '26)*

- Designed a per-frame **GP-BO** framework over **Cellpose** hyperparameters with a 6-D **ARD Matérn-1.5** kernel
- Introduced a **heteroscedastic sliding-window memory**, enabling **GP adaptation** under photobleaching drift
- Eliminated dense dataset annotation by integrating **Piecewise Cubic Hermite Interpolating Polynomials (PCHIP)**, generating a strict monotonic ground-truth trajectory from sparse ($\leq 1\%$) anchor frames annotation
- Formulated a biologically grounded **dead-zone loss** fxn (human undercount $\sim$12%), with asymmetric penalties
- Engineered an adaptive **dynamic-bounds expansion** protocol that seamlessly renormalizes GP memory to track extreme **SAM-regime** optima while preventing covariance collapse, improvising Cell Segmentation
- Achieved **TWCS** = 0.982, and **95.2%** in-band rate across **400** time-lapse frames (Pearson r=0.91, MAE = 2.17, MAPE = 5.62%); **Bland–Altman** bias +2.02 cells confirmed the undercounting of the reference annotation mask
- Extended the architecture to a fully **unsupervised, reference-free** variant utilizing **ParEGO** scalarization over intrinsic image heuristics for post-hoc **Pareto-front** analysis for better per frame Segmentation

Machine Learning based feature selection of markers for Alzheimer's disease

R&D Project | Proteomics Laboratory | Guide: Prof. Dr. Sanjeeva Srivastava *(Jan '24 - May'24)*

- Curated and analyzed available data from **BrainProt** and other research groups for curating large-scale dataset
- Implemented robust quality control, normalization, and preprocessing pipelines using **AnnData, Seurat, Scanpy**
- Implemented **differential expression analysis** to pinpoint differentially expressed genes, discovering markers
- Effectively utilized **Orange** for conducting PCA and markers verification and conducted model comparisons
- Executed implementations of **ANN, Logistic Regression, Naive Bayes, Random Forest, and SVM**
- **ANN** yielded highest precision rate of **95.334%**, via markers obtained for scrutinizing Alzheimer's disease

Curved Ionic Liquid Membranes for Enhancing Bio-Solvent Recovery

R&D Project | SLP IDP | Guide: Prof. Jhumpa Adhikari *(July '24 - Dec '24)*

- Calculated effective diffusion coefficients for Hexafluoroisopropanol ($0.0083 m^2/s$) and Methylcyclohexane using **Wilke-Chang equation**, incorporating membrane porosity and tortuosity factors to predict mass transport rates

- Selected and characterized ionic liquids ([EMIM][Tf_2N] & [HMIM][Tf_2N]) as green solvent-recovery media, evaluating key membrane parameters including liquid entry pressure (LEP ≈ 0.62 bar) and surface hydrophobicity
- Performed a failure-mode analysis revealing high solvent carryover under dead-end filtration (56% MCH retention), and proposed cross-flow filtration to mitigate fouling and concentration polarization based on simulation insights

Research Projects

Explainable Transfer Learning Benchmark for Neuro-Oncology MRI

Extended Research Project | GitHub | MedRxiv Preprint *(Updated 2026)*

- Overhauled an initial course prototype into an open-source transfer learning benchmark for brain MRI analysis
- Engineered custom restoration pipeline via **CLAHE** & unsharp masking to enhance soft-tissue boundary region
- Achieved peak perf with MobileNetV2 (**94.74%** acc, **0.994** ROC-AUC, **5.9 ms** latency) with **2.59M** parameters
- Applied **Grad-CAM** spatial attribution maps to validate clinical interpretability, to resolve black-box predictions and diagnostic heatmaps helps to provide radiologiss, interpretable visual evidence for automating the predictions

Predictive model generation for ligands binding to c-MYC G4 structures

Research Self Project *(July '23 - Dec '23)*

- Identified the basic descriptors for identification and classification of ligands binding to c-MYC G-4 structures
- Actively focused on the development and careful curation of a dataset containing over **2,000+** c-MYC G4 and **1,000+**ligand structures for ensuring comprehensive coverage of all possible ways and training the model
- Implemented robust quality control, normalization, and preprocessing pipelines using **AnnData, Seurat, Scanpy**
- Implemented **differential expression analysis** to pinpoint differentially expressed genes, discovering markers
- Utilized and expertly trained **k-Nearest Neighbors, Random Forest** machine learning models to effectively identify, distinguish, and classify complex structures with a **92.72%** accuracy above expected

Multiscale Modeling and Structural Stability Analysis of p53 Mutants

Course Project | MD Simulations for Materials Engineering | Guide: Prof. Ajay S. Panwar *(July '25 - Dec '25)*

- Successfully ported a **GROMACS** simulation workflow to **LAMMPS** to reproduce the dynamics of the **p53 DNA-binding** domain despite significant infrastructure incompatibilities in protein topology handling constraints
- Designed a robust MD pipeline (Minimization $\rightarrow$ NVT Equilibration $\rightarrow$ NPT Production) using a Truncated Octahedron box with TIP3P water and Monte-Carlo ion placement to mimic physiological salinity (0.15 M NaCl)
- Implemented a robust hybrid electrostatic model using Particle–Particle Particle–Mesh (PPPM) for long range accuracy (10^{-6}) and direct Coulomb interactions for the short range forces, optimized for a **32,810** atom system
- Achieved robust structural stability, evidenced by a backbone RMSD < 0.2 Å throughout the production run, validating the manual zinc-patching strategy and effectively mitigating the classical *flying ice cube* failure mode

Nanofluidic Modeling of Enhanced Mass Transport in Cell Membrane Channels

Course Project | MD Simulations for Materials Engineering | Guide: Prof. Ajay S. Panwar *(July '25 - Sept '25)*

- Developed a Non-Equilibrium Molecular Dynamics (NEMD) pipeline to model pressure-driven fluid transport through transmembrane nanopores, replicating the confined flow conditions & characteristic of cellular channels
- Quantified a **$\sim$1.8$\times$ enhancement in mass transport rates**($J \approx 2.9$ vs. 1.6) in hydrophobic channels relative to hydrophilic counterparts, providing mechanistic insight into high-efficiency transport observed in cell aquaporins.
- Identified the emergence of a **slip boundary condition** in hydrophobic confinement ($\epsilon_{12} = 0.2$), where reduced interfacial friction enables rapid flow, challenging the classical no-slip assumptions in the biological fluid mechanics

Elevating Spectra Classification through Advanced Machine Learning Techniques

R&D Project | CH 433 | Prof. Amber Jain and Prof. Nand Kishore *(July '23 - Nov '23)*

- Utilized the **k-nearest neighbors** algorithm for precise classification, while simultaneously applying many machine learning techniques to classify the infrared spectra into "carbonyl" and "non-carbonyl" categories
- Utilized **PCA** and **SMOTE** for data simplification, focusing on the chemical element property correlations
- Employed the **ASAP-MS** spectroscopy and the **FTIR** spectroscopy data for versatile characterisation of samples
- Expertly applied the **k-NN, SVM, and LDA** as the classifiers for pattern recognition achieved **93.4%** accuracy

Entrepreneurial Experience

Co-Founder: ShellCrete
Global Scientific Challenge | Imperial College London | Our Website *(July '26)*

- Co-founded a 5-person venture selected for **Imperial College Global Challenge**, converting shell biowaste into **ShellCrete**, a partial cement replacement targeting the cement industry's **5–8%** share of global CO_2 emissions
- Researched **20+ peer-reviewed studies**, defining an evidence-based 5–10% cement-substitution formula and 900–950°C calcination protocol projected to cut upstream limestone demand by **43–122 kg per tonne of binder**
- Built **ShellPass**, a material-traceability platform for mapping waste from collection to processing to end product
- Designed and ran a **1000+ respondent market-validation survey**, finding out **84.4%** of the respondents would trust a certified shell-based product, to validate the demand assumptions behind the venture's business case
- Co-authored and presented our investor pitch deck to competition judges and VC's, translating the technical case into a business model and 4-year scaling roadmap (now UK pilot → international expansion by 2030)

Technical Projects

Multi-PDF Querying using LLMs
Course Project | *Speech, NLP and the Web* | Prof. Pushpak Bhattacharyya *(Jan '24 - Jun '24)*

- **Developed** a python **GUI** based application using **LangChain, Streamlit and HuggingFace LLMs**
- Used **PyPDF2** for pdfs to mono-string, **CharacterTextSplitter** for chunks & **instructor-xl embedding**
- Utilized a non-persistent **FAISS** vector store in **conversation-chain** using the free **Google's flan-t5-xxl LLM**
- **Reused** the chat history using **ConversationBufferMemory** of LangChain & Streamlit's **session variables**

Positions of Responsibility

UG Representative | Chemistry Department, IIT Bombay *(May '24 - May '25)*

- Selected through an exceptionally rigorous selection process standing out among candidates in department
- Representing the entire chemistry department, ensuring their diverse needs and concerns are effectively addressed

Institute Student Mentor | Student Mentor Program, IIT Bombay *(July '24 - July '26)*

- Selected amongst **435** students in 4th and 5th year applicants across the institute after rigorous inter-views and peer-reviews on the basis of balanced academics, extracurricular and mentoring skills to guide students
- Mentoring **12** freshmen with their academic & co-curricular pursuits and fostering their all-round personal growth

Department Placement Coordinator | Placement Team, IIT Bombay *(July '24 - July '25)*
Assisting job placements for 2100+ students, bridging the gap between the placement cell and students effectively

- Devised efficient PoA for **10+** upskilling events, aligning students with industry standards in many portfolios
- Streamlined company pitching with preferences from **30+** students, ensuring max placements of students
- Unanimously elected to represent **30+** department students, contributing to success of Institute Placement Team

Extracurriculars

- Organized Chemistry Department Convocation Day for the past **2 years**, managing all logistics *[2022 & '23]*
- Serving as **class representative** of my batch for the past **2 years**, acting as the voice of my batch *[2022 & '23]*
- Mentored a batch of **8** students as **WIDS mentor**, delivering **in-depth** instruction in **ML** concepts *[2023]*
- Mentored batch of **20+** students as **SoC mentor**, delivering concepts in **supply chain optimization** *[2023]*
- Secured **1st** position in school and was awarded **Gold** Medal in **National Science Olympiad** *[2016]*
- Completed a year-long professional **Weight-Lifting** training under **National Sports Association** *[2022]*
- Coordinated with Abhyuday to organize Versova Beach cleanup, fostering environmental awareness *[2022]*
- Secured **1st** position in **Chem Detective GC** by **Chemistry Club**, for Hostel 2 [2023]